

Elementary Matrix and a Method For Finding A^{-1}

Definition :- Matrices A and B are said to be row equivalent if either (hence each) can be obtained from other by a sequence of elementary row operations.

Elementary Matrix:- Matrix E is called ~~an~~ elementary matrix if it can be obtained from an identity matrix by performing a single elementary row operation.

Example 1

$$\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$$

↓
Multiplying
2nd row of
 I_2 by -3

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

↓
Interchanging 2nd
and 4th row of I_4

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Add $3R_3$ of I_3 to
1st row.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ Multiply R_1 of I_3
by 1.

Theorem 1.5.1 (Row operation by Matrix Multiplication)

Row operations on any matrix can also be obtained
by elementary matrix.

Example: 2 (Using Elementary Matrix)

$$A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 1 & 4 & 4 & 0 \end{bmatrix}$$

Consider elementary matrix: $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$ → obtain
by $R_3 + 3R_1$
on I_3

$$EA = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 3 & 6 \\ 4 & 4 & 10 & 9 \end{bmatrix}$$

→ This is precisely the matrix
that result when we
add 3 times the first
row of A to third row.

→ Inverse

Row operations on I
that produces E

Row operations on E
that reproduces I

1. Multiply row i by $c \neq 0$

Multiply row i by $1/c$

↳ on I
matrix

2. Interchange rows i and j

Interchange row i and j

3. Add c times row i to row j

Add $-c$ times row i to row j

↳ This tells that row operations
are inverse of each other.

Example-3 (Row Operations and inverse row ops)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow[\text{row by 7}]{\text{multiply 2nd}} \begin{bmatrix} 1 & 0 \\ 0 & 7 \end{bmatrix} \xrightarrow[\text{row by } \frac{1}{7} \text{ (reverse op)}]{\text{x-ply second}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow[\text{Int interchange 1st \& 2nd row}]{\downarrow} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \xrightarrow[\text{Int interchange 1st \& 2nd row (reverse op)}]{\downarrow} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow[\text{add 5 times second row to the 1st}]{\downarrow} \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix} \xrightarrow[\text{add -5 times 2nd row to 1st}]{\downarrow} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{I, -5R}$$

key point: Every elementary matrix is invertible
and inverse is also invertible.

Theorem 1.5.3 (Equivalent statements)

If A is $(n \times n)$ / square matrix, then following statements are equivalent, that is, all true or all false.

1. A is invertible
2. $AX=0$ has only trivial solution
3. The reduced row echelon form of A is I_n
4. A is expressible as product of elem. matrices

Suppose if machine (A) is invertible (perfect).
It never loss info. It takes input x and given output o . The only way output becomes 0 is when input was 0.

Inversion Algo.

To find inverse of invertible matrix A , find a seq. of elementary row operations that reduce A to identity and then perform same sequence of operations on I_n to obtain A^{-1} .

Example 4:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \quad \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\begin{array}{l} R_1 - 2R_2 \\ R_3 + 2R_2 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & +1 & +5 & -2 & -1 \end{array} \right] \begin{array}{l} \\ -R_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & +13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \begin{array}{l} R_1 - 9R_3 \\ R_2 + 3R_3 \end{array}$$

so

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ +13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \quad \text{Ans}$$

Example: 5 (Showing that a matrix is not inv.)

$$A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ -1 & 2 & 5 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & -2 & 1 & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 + R_1 \end{array}$$

$$-\frac{R_2}{8} \left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & 1 & 9/8 & 1/4 & -1/8 & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -1/4 & -1/2 & 3/4 & 0 \\ 0 & 1 & 9/8 & 1/4 & -1/8 & 0 \\ 0 & 0 & 0 & -1 & 1 & 1 \end{array} \right]$$

↓

Since we have obtained a row of zero on left
so A is not invertible.

Example: 6 (Analyze homogenous system)

$$\begin{aligned} a) \quad & x_1 + 2x_2 + 3x_3 = 0 \\ & 2x_1 + 5x_2 + 3x_3 = 0 \\ & x_1 + 8x_3 = 0 \end{aligned} \rightarrow \text{This is the same matrix as of example 4}$$

From theorem 1.5.3 a system has only trivial iff coefficient matrix is invertible. In example 4 we have seen that the matrix is invertible so the system has trivial solution.

$$\begin{aligned} b) \quad & x_1 + 6x_2 + 4x_3 = 0 \\ & 2x_1 + 4x_2 - x_3 = 0 \\ & -x_1 + 2x_2 + 5x_3 = 0 \end{aligned} \rightarrow \text{same matrix as of example 5}$$

From theorem 1.5.3 a system has trivial sol. iff coeff. matrix is invertible. In example 5 we have seen that matrix is not invertible so system has non trivial solution.

Exercise Set 1.5

Question: 01

Determine whether ~~elementary~~ given matrix is elementary.

a-

$$\begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix} \rightarrow \text{Yes elem. and obtained by} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix} R_2 - 5R_1$$

b-

$$\begin{bmatrix} -5 & 1 \\ 1 & 0 \end{bmatrix}$$

→ No, because it can't be obtained by applying single ~~element~~ row op on I_2

c-

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

→ No, can't be obtain by applying Single row ope. on I_3

d-

$$\begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

→ No, can't be obtain by applying single row operation on I_4 .

Question: 02

Determine whether given matrix is elem. or not

a-

$$\begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{bmatrix}$$

→ Yes, and obtain $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ by $\sqrt{3}R_2$

b-

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

→ Yes, obtain $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ by $R_1 \leftrightarrow R_3$

c-

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 9 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[\text{by}]{\text{Yes, obtain}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_2 + 9R_3$$

d-

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \rightarrow \text{No, because can't be obtained by applying on single row op on } I_3$$

Question 3-4

Find a row op and corresponding elementary matrix that will restore given elementary matrix to identity matrix.

Question: 03

a-

$$\begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} \xrightarrow{\text{obtained by}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{by } R_1 - 3R_2$$

1

↳ Row operation.

To restore Identity.

$$\begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad R_1 + 3R_2$$

To Find elementary matrix.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \quad R_1 + 3R_2$$

↳ Elem. Matrix

b-

$$\begin{bmatrix} -7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Row operation.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} -7R_1 \\ \rightarrow \text{Row oper.} \end{matrix}$$

To Restore Identity

$$\begin{bmatrix} -7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} -\frac{1}{7}R_1$$

To Find Elem.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -\frac{1}{7} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} -\frac{1}{7}R_1$$

$\rightarrow$ Elementary Matrix.

c-

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Row oper.}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix} \begin{matrix} \rightarrow R_3 - 5R_1 \\ \downarrow \text{row oper.} \end{matrix}$$

To restore identity

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_3 + 5R_1$$

To Find elementary matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & 0 & 1 \end{bmatrix} \rightarrow \text{Elementary Matrix}$$

d-

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Row Oper.}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} R_1 \leftrightarrow R_3 \\ \downarrow \\ \text{Row Oper.} \end{array}$$

To Restore Identity

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} R_1 \leftrightarrow R_3$$

To Find Elem.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} R_1 \leftrightarrow R_3 \\ \rightarrow \text{Element Matrix.} \end{array}$$

Question 4:

a-

$$\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \xrightarrow{\text{Row Oper}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_2 - 3R_1} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{array}{l} R_2 - 3R_1 \\ \downarrow \\ \text{Row Operation} \end{array}$$

To Store Identity.

$$\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} R_2 + 3R_1$$

To Find elementary.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{array}{l} R_2 + 3R_1 \\ \rightarrow \text{Elementary matrix} \end{array}$$

b-

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \xrightarrow{\text{Row op}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{3R_3 \rightarrow \text{Row op}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

To Restore Identity.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\frac{1}{3}R_3}$$

To Find elem. matrix.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \xrightarrow{\frac{1}{3}R_3} \text{Elem. Matrix.}$$

c-

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{Row op.}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_4}$$

To Restore Idem.

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_4}$$

To Find elem.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_4} \text{Elem. Matrix.}$$

d-

Row oper

$$\begin{bmatrix} 1 & 0 & -\frac{1}{7} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Row oper}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} R_1 + \frac{1}{7}R_3 \\ \text{Row oper} \end{matrix}} \begin{bmatrix} 1 & 0 & -\frac{1}{7} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

To restore Idem:

$$\begin{bmatrix} 1 & 0 & -\frac{1}{7} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{By } R_1 + \frac{1}{7}R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

To Find elem. Matrix.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} R_1 + \frac{1}{7}R_3 \\ \text{Elem. Matrix} \end{matrix}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Question : 05-06

An elem. matrix E and matrix A are given identify row operation corresponding to E and verify that the product EA is result from applying row operation to A.

Question : 5

a-

$$E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$$

Row oper of E

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Product EA

$$EA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$$

$$EA = \begin{bmatrix} 0+3 & 0-6 & 0-6 & 0-6 \\ -1+0 & -2-0 & 5-0 & -1-0 \end{bmatrix}$$

$$EA = \begin{bmatrix} 3 & -6 & -6 & -6 \\ -1 & -2 & 5 & -1 \end{bmatrix}$$

verify by applying $R_1 \leftrightarrow R_2$ on A

$$A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & -6 & -6 & -6 \\ -1 & -2 & 5 & -1 \end{bmatrix} R_1 \leftrightarrow R_2$$

↳ exactly = EA

b-

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}, A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$$

Row operation on E

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} R_3 - 3R_2$$

Product EA

$$EA = \begin{bmatrix} 2+0+0 & -1-0+0 & 0-0+0 & -4+0+0 & -4+0+0 \\ 0+1+0 & -0-3+0 & 0-1+0 & -0+5+0 & -0+3+0 \\ 0-3+2 & -0+9+0 & 0+3+1 & -0-15+3 & -0-9+1 \end{bmatrix}$$

$$EA = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ -1 & 9 & 4 & -12 & -10 \end{bmatrix}$$

verify by applying $R_3 - 3R_2$ on A

$$A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ -1 & 9 & 4 & -12 & -10 \end{bmatrix} \begin{matrix} \\ R_3 - 3R_2 \\ \end{matrix}$$

$\rightarrow \text{exactly} = EA$

C-

$$E = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$\rightarrow R_1 + 4R_3$

Method will be same.

Question : 6

a-

$$E = \begin{bmatrix} -6 & 0 \\ 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$$

$\rightarrow -6R_1$

Same method will be followed as of Q: 5

b-

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$$

Row operation on E

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_2 - 4R_1$$

Product EA :

$$EA = \begin{bmatrix} 2+0+0 & -1-0+0 & 0-0+0 & -4+0+0 & -4+0+0 \\ -8+1+0 & 4-3+0 & -0-1+0 & 16+5+0 & 16+3+0 \\ 0+0+2 & -0-0+0 & 0-0+1 & -0+0+3 & -0+0+1 \end{bmatrix}$$

$$EA = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ -7 & 1 & -1 & 21 & 19 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$$

Verify by applying $R_2 - 4R_1$ on A

$$A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ -7 & 1 & -1 & 21 & 19 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix} \quad R_2 - 4R_1$$

$\hookrightarrow \text{Exactly} = EA$

C-

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$\hookrightarrow 5R_2$

same method will be followed.

Question: 7-8

Use following matrices and find an elementary matrix E that satisfies the stated equations

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}, B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}$$

$$D = \begin{bmatrix} 8 & 1 & 5 \\ -6 & 21 & 3 \\ 3 & 4 & 1 \end{bmatrix}$$

$$F = \begin{bmatrix} 8 & 1 & 5 \\ 8 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}$$

Question: 7

a-

$$EA = B$$

→ compare A and B and we can see B is obtain by applying $(R_1 \leftrightarrow R_3)$ on A

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad R_1 \leftrightarrow R_3$$

$$\downarrow$$

so $E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

b-

$EB = A$ → compare B and A and we can see A is obtain by applying $(R_1 \leftrightarrow R_3)$ on B

c-

$$EA = C$$

compare A and C we can see C is obtain by applying $(R_3 - 2R_1)$ on A

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad R_3 - 2R_1$$

$$\text{so } E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

d-

$$EC = A$$

compare C and A, we can see that A is obtained by applying $(R_3 + 2R_1)$ on C

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \quad R_3 + 2R_1$$

so $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

Question: 8

a-

$$EB = D$$

compare B and D and we can see D is obtained by applying $(-3R_2)$ on B

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{-3R_2} E$$

b-

$$ED = B$$

compare D and B and we can see B is obtained by applying $-\frac{1}{3}R_2$ on D

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{-\frac{1}{3}R_2} E$$

c-

$$EB = F$$

compare B and F and we can see F is obtain by applying $R_2 + 2R_3$ on B.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 + 2R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow E$$

d-

$$EF = B$$

compare F and B and we can see B is obtain by applying $R_2 - 2R_3$ on F.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 - 2R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow E$$

Question : 9-10

Use theorem 1.4.5 and then use inversion algorithm to find A^{-1} if exist

Question : 9

b-

$$A = \begin{bmatrix} 2 & -4 \\ -4 & 8 \end{bmatrix}$$

Normal Method

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

$\det A$

$$\det A = (2 \times 8) - (-4)(-4)$$

$$\det A = 0 \rightarrow$$

Inversion Method.

$$\left[\begin{array}{cc|cc} 2 & -4 & 1 & 0 \\ -4 & 8 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cc|cc} 1 & -2 & \frac{1}{2} & 0 \\ -4 & 8 & 0 & 1 \end{array} \right] \xrightarrow{\frac{1}{2} R_1}$$

Inverse not
exist as $\det A = 0$

$$\left[\begin{array}{cc|cc} 1 & -2 & \frac{1}{2} & 0 \\ 0 & 0 & 2 & 1 \end{array} \right] R_2 + 4R_1$$

Since ~~left side~~ we have
obtained row of 0 on left
side so inverse not exist

a-

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix}$$

→ same method will be
followed.

Question : 10-

a-

b- same as part
(a).

$$A = \begin{bmatrix} -1 & -5 \\ 3 & -16 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

$$\det A = (1)(-16) - (3)(-5)$$

$$\det A = -1$$

$$\text{adj } A = \begin{bmatrix} 16 & 5 \\ -3 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 16 & 5 \\ -3 & -1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} +16 & -5 \\ +3 & -1 \end{bmatrix}$$

$$\left[\begin{array}{cc|cc} 1 & -5 & 1 & 0 \\ 3 & -16 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cc|cc} 1 & -5 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array} \right] R_2 - 3R_1$$

$$\left[\begin{array}{cc|cc} 1 & -5 & 1 & 0 \\ 0 & 1 & 3 & -1 \end{array} \right] -R_2$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 16 & -5 \\ 0 & 1 & 3 & -1 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} 16 & -5 \\ 3 & -1 \end{bmatrix}$$

Question: 11-12-

use inversion algorithm to find inverse of matrix
(if exist)

Question 11-

a-

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 9 & -5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \begin{array}{l} R_1 - 2R_2 \\ R_3 + 2R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 9 & -5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & +5 & -2 & -1 \end{array} \right] -R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & +13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

b-

Same procedure will be used:

$$A = \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

Question: 12

a

$$\begin{bmatrix} \frac{1}{5} & \frac{1}{5} & -\frac{2}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{10} \\ \frac{1}{5} & -\frac{4}{5} & \frac{1}{10} \end{bmatrix}$$

b-

same method will be followed:

$$\left[\begin{array}{ccc|ccc} \frac{1}{5} & \frac{1}{5} & -\frac{2}{5} & 1 & 0 & 0 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{10} & 0 & 1 & 0 \\ \frac{1}{5} & -\frac{4}{5} & \frac{1}{10} & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 2 & 2 & 1 & 0 & 10 & 0 \\ 2 & -8 & 1 & 0 & 0 & 10 \end{array} \right] \begin{array}{l} R_1 (5) \\ R_2 \\ R_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & 0 & 5 & -10 & 10 & 0 \\ 0 & -10 & 5 & -10 & 0 & 10 \end{array} \right] \begin{array}{l} R_2 - 2R_1 \\ R_3 - 2R_1 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -10 & 5 & -10 & 0 & 10 \\ 0 & 0 & 5 & -10 & 10 & 0 \end{array} \right] R_2 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 1 & 0 & -1 \\ 0 & 0 & 5 & -10 & 10 & 0 \end{array} \right] \begin{array}{l} -R_1 \\ 10 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{3}{2} & 4 & 0 & 1 \\ 0 & 1 & -\frac{1}{2} & 1 & 0 & -1 \\ 0 & 0 & 5 & -10 & 10 & 0 \end{array} \right] \begin{array}{l} R_3 - R_2 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{3}{2} & 4 & 0 & 1 \\ 0 & 1 & -\frac{1}{2} & 1 & 0 & -1 \\ 0 & 0 & 1 & -2 & 2 & 0 \end{array} \right] \begin{array}{l} \frac{1}{5} R_3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 3 & 1 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -2 & 2 & 0 \end{array} \right] \begin{array}{l} R_2 + \frac{1}{2} R_3 \end{array}$$

$$A^{-1} = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & -1 \\ -2 & 2 & 0 \end{bmatrix} \text{ Ans}$$

Question : 13-18

Use inversion algorithm to find the inverse of matrix (if exist)

14-

$$\begin{bmatrix} \sqrt{2} & 3\sqrt{2} & 0 \\ -4\sqrt{2} & \sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{3} & 3\sqrt{2} & 0 & | & 1 & 0 & 0 \\ -4\sqrt{2} & \sqrt{2} & 0 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 0 & | & \frac{1}{\sqrt{2}} & 0 & 0 \\ -4 & 1 & 0 & | & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} \frac{1}{\sqrt{2}} R_1 \\ \frac{1}{\sqrt{2}} R_2 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 0 & | & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 13 & 0 & | & \frac{4}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix} R_2 + 4R_1$$

$$\begin{bmatrix} 1 & 3 & 0 & | & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 1 & 0 & | & \frac{4}{13\sqrt{2}} & \frac{1}{13\sqrt{2}} & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix} \frac{1}{13} R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{13\sqrt{2}} & \frac{-3}{13\sqrt{2}} & 0 \\ 0 & 1 & 0 & \frac{4}{13\sqrt{2}} & \frac{1}{13\sqrt{2}} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$A^{-1} = \left[\begin{array}{ccc} \frac{1}{13\sqrt{2}} & \frac{-3}{13\sqrt{2}} & 0 \\ \frac{4}{13\sqrt{2}} & \frac{1}{13\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{array} \right] \text{ Ans}$$

$$13 \quad | \quad 15 \quad | \quad 18 \quad | \quad 16 \quad | \quad 17$$

same procedure will be followed for all these.

Question: 19-20

Find inverse of each of following matrix (4×4) where k_1, k_2, k_3, k_4 and k are all non-zero.

Question: 19-

a.

$$\begin{bmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & 0 \\ 0 & 0 & 0 & k_4 \end{bmatrix}$$

→ matrix is diagonal. So inverse of diagonal matrix is simply a diagonal matrix where each entry is reciprocal of original

$$A^{-1} = \begin{bmatrix} \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & \frac{1}{k_3} & 0 \\ 0 & 0 & 0 & \frac{1}{k_4} \end{bmatrix} \text{ Ans}$$

b-

$$\begin{bmatrix} k & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & k & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

matrix consist of two 2×2 block along diagonal

$$\begin{bmatrix} k & 1 \\ 0 & 1 \end{bmatrix} \text{ and } \begin{bmatrix} k & 1 \\ 0 & 1 \end{bmatrix}$$

convert 1st block to inverse.

$$\left[\begin{array}{cc|cc} k & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} k & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 \end{array} \right] R_1 - R_2$$

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{k} & -\frac{1}{k} \\ 0 & 1 & 0 & 1 \end{array} \right] \frac{1}{k} R_2$$

since both block are same
so inverse will be.

$$A^{-1} = \begin{bmatrix} \frac{1}{k} & -\frac{1}{k} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{k} & -\frac{1}{k} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Question: 20-

a-

$$\begin{bmatrix} 0 & 0 & 0 & k_1 \\ 0 & 0 & k_2 & 0 \\ 0 & k_3 & 0 & 0 \\ k_4 & 0 & 0 & 0 \end{bmatrix}$$

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & k_3 & 0 & 0 & 0 & 0 & 1 & 0 \\ R_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} k_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & k_3 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & R_1 & 1 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} R_1 \leftrightarrow R_4 \\ R_2 \leftrightarrow R_3 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{k_4} \\ 0 & 1 & 0 & 0 & 0 & 0 & \frac{1}{k_3} & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & 0 & 1 & \frac{1}{k_1} & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} \frac{1}{k_4} R_1 \\ \frac{1}{k_3} R_2 \\ \frac{1}{k_2} R_3 \\ \frac{1}{k_1} R_4 \end{array}$$

$$\text{So } A^{-1} = \left[\begin{array}{cccc} 0 & 0 & 0 & \frac{1}{k_4} \\ 0 & 0 & \frac{1}{k_3} & 0 \\ 0 & \frac{1}{k_2} & 0 & 0 \\ \frac{1}{k_1} & 0 & 0 & 0 \end{array} \right] \quad \text{Ans}$$

Q.20 (b)

$$\left[\begin{array}{cccc} k & 0 & 0 & 0 \\ 1 & k & 0 & 0 \\ 0 & 1 & k & 0 \\ 0 & 0 & 1 & k \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} k & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & k & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & k & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} k & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & k & 0 & 0 & -\frac{1}{k} & 1 & 0 & 0 \\ 0 & 1 & k & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array} \right] \quad R_2 \leftarrow R_2 - \frac{1}{k} R_1$$

$$\begin{bmatrix} k & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & k & 0 & 0 & -\frac{1}{k} & 1 & 0 & 0 \\ 0 & 0 & k & 0 & \frac{1}{k^2} & -\frac{1}{k} & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{bmatrix} \quad R_3 - \frac{1}{k} R_2$$

$$\begin{bmatrix} k & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & k & 0 & 0 & -\frac{1}{k} & 1 & 0 & 0 \\ 0 & 0 & k & 0 & \frac{1}{k^2} & -\frac{1}{k} & 1 & 0 \\ 0 & 0 & 0 & k & -\frac{1}{k^3} & \frac{1}{k^2} & -\frac{1}{k} & 1 \end{bmatrix} \quad R_4 - \frac{1}{k} R_3$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{k^2} & \frac{1}{k} & 0 & 0 \\ 0 & 0 & 1 & 0 & +\frac{1}{k^3} & -\frac{1}{k^2} & \frac{1}{k} & 0 \\ 0 & 0 & 0 & 1 & -\frac{1}{k^4} & \frac{1}{k^3} & -\frac{1}{k^2} & \frac{1}{k} \end{array} \right] \quad \begin{array}{l} \frac{1}{k} R_1 \\ \frac{1}{k} R_2 \\ \frac{1}{k} R_3 \\ \frac{1}{k} R_4 \end{array}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{k} & 0 & 0 & 0 \\ -\frac{1}{k^2} & \frac{1}{k} & 0 & 0 \\ +\frac{1}{k^3} & -\frac{1}{k^2} & \frac{1}{k} & 0 \\ -\frac{1}{k^4} & \frac{1}{k^3} & -\frac{1}{k^2} & \frac{1}{k} \end{bmatrix}$$

Question: 21-22 ^{if any}

find all values of c for which the given matrix is invertible.

21-

$$A = \begin{bmatrix} c & c & c \\ 1 & c & c \\ 1 & 1 & c \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & c \\ 1 & c & c \\ c & c & c \end{bmatrix} \quad R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 1 & c \\ 0 & c-1 & 0 \\ c & c & c \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 1 & c \\ 0 & c-1 & 0 \\ 0 & 0 & c-c^2 \end{bmatrix} \xrightarrow{R_3 - cR_1}$$

Invertible if $c-1 \neq 0$ and $c-c^2 \neq 0$

$$\begin{array}{l|l} c-1 \neq 0 & c-c^2 \neq 0 \\ c \neq 1 & c(1-c) \neq 0 \\ & c \neq 0, 1 \neq c \end{array}$$

Conclusion: A is invertible for all values of c except $c=0$ and $c=1$

22-

$$\begin{bmatrix} c & 1 & 0 \\ 1 & c & 1 \\ 0 & 1 & c \end{bmatrix}$$

$$\begin{bmatrix} 1 & c & 1 \\ c & 1 & 0 \\ 0 & 1 & c \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & c & 1 \\ 0 & 1-c^2 & -c \\ 0 & 1 & c \end{bmatrix} \xrightarrow{R_2 - cR_1}$$

$$\begin{bmatrix} 1 & c & 1 \\ 0 & 1 & c \\ 0 & 1-c^2 & -c \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & c & 1 \\ 0 & 1 & c \\ 0 & 0 & c(-2+c^2) \end{bmatrix}$$

Invertible if

$$c(-2+c^2) \neq 0$$

$$c \neq 0, c^2 - 2 \neq 0$$

$$c \neq 0, c^2 \neq 2$$

$$c \neq 0, c \neq \pm\sqrt{2}$$

$$-c - (1-c^2)(c)$$

$$-c - c(1-c^2)$$

$$-c - c + c^3$$

$$-2c + c^3$$

$$c(-2+c^2)$$

So A is invertible if $c \neq 0$ and $c \neq \pm\sqrt{2}$